



University of Computer Studies, Yangon (UCSY)

Introduction

IS-212 (Data and Knowledge Mining)

Dr. Hsu Myat Mo

Faculty of Information Science, UCSY



References

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- Data Mining Concepts and Techniques, Jiawei Han, Micheline Kamber, Jian Pei. 4th Edition, Morgan Kaufmann.
- Introduction to Data Mining, Pang-Ning Tan, Michael Steinbach, Vipin Kumar, Addison Wesley, 2006.
- Data Mining – Practical Machine Learning Tools and Techniques (2nd Ed.), Ian H. Witten and Eibe Frank, Morgan Kaufmann, 2005.
- Data Mining with R – Learning with Case Studies, Torgo, Luís, CRC Press, Taylor and Francis Group, LLC, 2011



Course Objectives

- To explain the workflow of data mining techniques
- To remember how to manipulate data mining techniques with Python and R-Programming
- To explain how to evaluate the experimental results of data mining techniques
- To express the visualization of the experimental results with Python and R-Programming



Outlines

- Introduction
- Data Mining



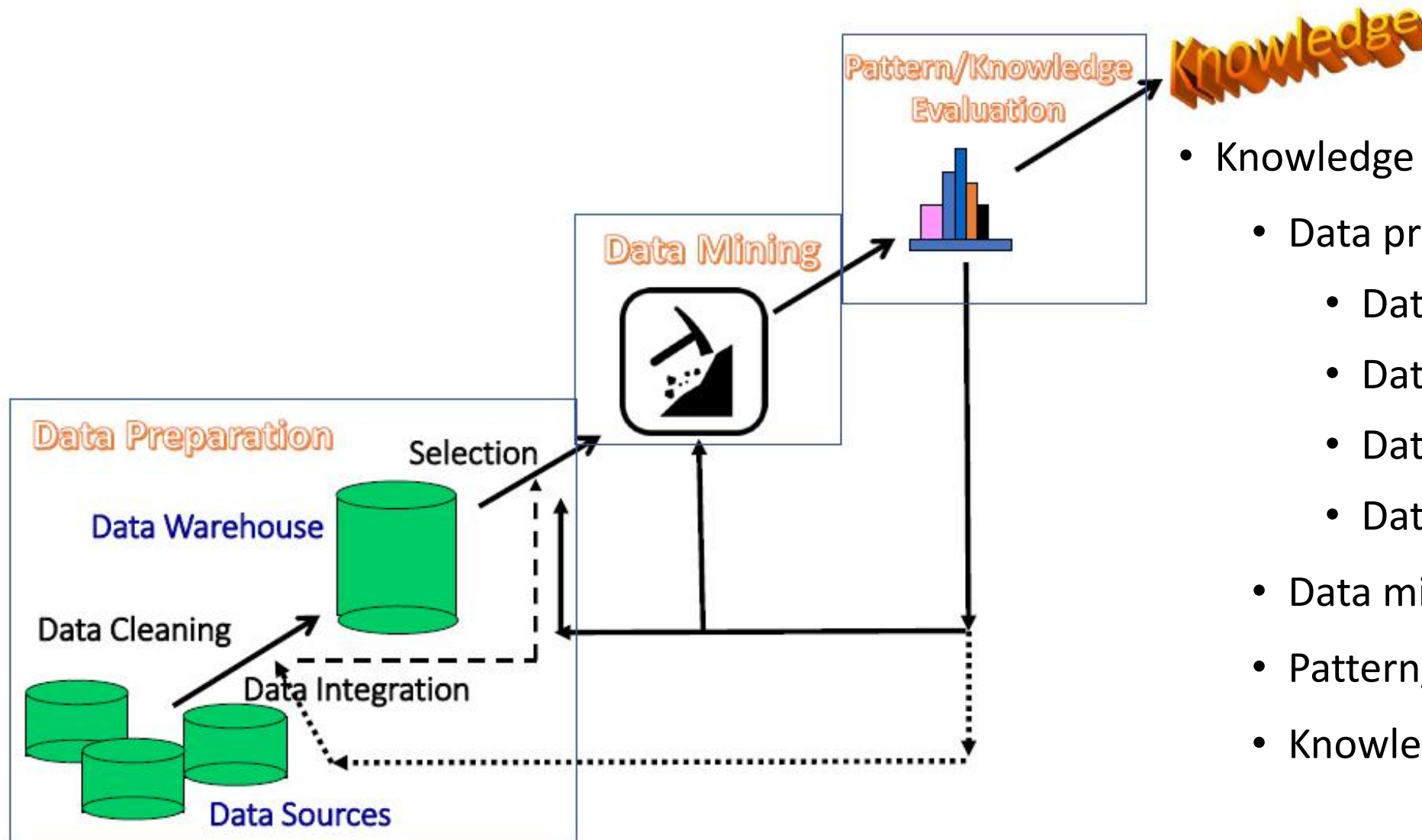
Data Mining

- **Data mining** is the process of semi-automatically analyzing large databases to find useful patterns
 - Similar goals to machine learning, but on very large volumes of data
- Part of the larger area of knowledge discovery in databases (KDD)
- Some types of knowledge can be represented as rules
- More generally, knowledge is discovered by applying **machine learning techniques** on past instances of data, to form a model
 - Model is then used to make predictions for **new instances**

➤ Data mining is the process of discovering interesting patterns, models and knowledge from large amounts of data.



Data Mining: An Essential Step in Knowledge Discovery



- Knowledge Discovery Process
 - Data preparation
 - Data cleaning
 - Data integration
 - Data transformation
 - Data selection
 - Data mining
 - Pattern/model evaluation
 - Knowledge presentation



Knowledge Discovery from Data(KDD) Process

- **Data cleaning** - Remove noise and inconsistent data
- **Data Integration** - Multiple data sources may be combined
- **Data Selection** - Data relevant to the analysis task are retrieved from the database
- **Data Transformation** - Transformed and consolidated into appropriate forms by aggregation
- **Data Mining** - Intelligent methods to extract data patterns
- **Pattern Evaluation** - Interesting patterns representing knowledge based on interestingness measures
- **Knowledge Presentation** - Visualization and knowledge representation techniques



Data Mining : On What Kinds of Data?

- Database-oriented data sets and applications
 - Relational database, data warehouse, transactional database
- Advanced data sets and advanced applications
 - Data streams and sensor data
 - Time-series data, temporal data, sequence data (incl. bio-sequences)
 - Structure data, graphs, social networks and multi-linked data
 - Object-relational databases
 - Heterogeneous databases and legacy databases
 - Spatial data and spatiotemporal data
 - Multimedia database
 - Text databases
 - The World-Wide Web



What Kinds of Patterns Can Be Mined?



Types of Data Mining Tasks

- **Prediction** based on past history
 - Predict if a credit card applicant poses a good credit risk, based on some attributes (income, job type, age, ..) and past history
 - Predict if a pattern of phone calling card usage is likely to be fraudulent
- Some examples of prediction mechanisms:
 - **Classification**
 - Items (with associated attributes) belong to one of several classes
 - **Training instances** have attribute values and classes provided
 - Given a new item whose class is unknown, predict to which class it belongs based on its attribute values
 - **Regression** formulae
 - Given a set of mappings for an unknown function, predict the function result for a new parameter value



Types of Data Mining Tasks

➤ Descriptive Patterns

➤ Associations

- Find books that are often bought by “similar” customers. If a new such customer buys one such book, suggest the others too.
- Associations may be used as a first step in detecting **causation**
 - E.g., association between exposure to chemical X and cancer,

➤ Clusters

- E.g., typhoid cases were clustered in an area surrounding a contaminated well
- Detection of clusters remains important in detecting epidemics

- Descriptive mining tasks characterize properties of the data in a target data set.
- Predictive mining tasks perform induction on the current data in order to make predictions.



Data Mining Functions: Generalization

- Information integration and data warehouse construction
 - Data cleaning, transformation, integration, and multidimensional data model
- Data cube technology
 - Scalable methods for computing (i.e., materializing) multidimensional aggregates
 - OLAP (online analytical processing)
- Multidimensional concept description: Characterization and discrimination
 - Generalize, summarize, and contrast data characteristics, e.g., dry vs. wet region



Class/Concept Description: Characterization and Discrimination

- Data entries can be associated with classes or concepts
- class/concept descriptions can be derived using
 - (1) *data characterization*, by summarizing the data of the class under study (often called the target class) in general terms, or
 - (2) *data discrimination*, by comparison of the target class with one or a set of comparative classes (often called the contrasting classes), or
 - (3) *both* data characterization and discrimination



Class/Concept Description: Characterization and Discrimination

- **Data characterization** is a summarization of the general characteristics or features of a target class of data.
- **Data discrimination** is a comparison of the general features of the target class data objects against the general features of objects from one or multiple contrasting classes.

Other Types of Mining



- **Text mining**: application of data mining to textual documents
- **Sentiment analysis**
 - E.g., learn to predict if a user review is positive or negative about a product
- **Information extraction**
 - Create structured information from unstructured textual description or semi-structured data such as tabular displays
- **Entity recognition** and **disambiguation**
 - E.g., given text with name “Michael Jordan” does the name refer to the famous basketball player or the famous ML expert
- **Knowledge graph**
 - Can be constructed by information extraction from different sources, such as Wikipedia

Other Mining Various Kinds of Knowledge



- Multidimensional Data Summarization
- Mining Frequent Patterns, Associations, and Correlations
- Classification and Regression for Predictive Analysis
- Cluster Analysis
- Deep Learning
- Outlier Analysis
- Are All Mining Results Interesting?

Evaluation of Knowledge

- Are all mined knowledge interesting?
 - One can mine tremendous amount of “patterns”
 - Some may fit only certain dimension space (time, location, ...)
 - Some may not be representative, may be transient, ...
- Evaluation of mined knowledge → directly mining only interesting knowledge?
 - Descriptive vs. predictive
 - Coverage
 - Typicality vs. novelty
 - Accuracy
 - Timeliness
 - ...





What makes a pattern interesting?

- A pattern is *interesting* if it is
 - (1) *easily understood* by humans,
 - (2) *valid* on new or test data with some degree of *certainty*,
 - (3) potentially *useful*, and
 - (4) *novel*.
- A pattern is also interesting if it validates a hypothesis that the user *sought to confirm*.
- An interesting pattern represents *knowledge*.

What makes a pattern interesting?



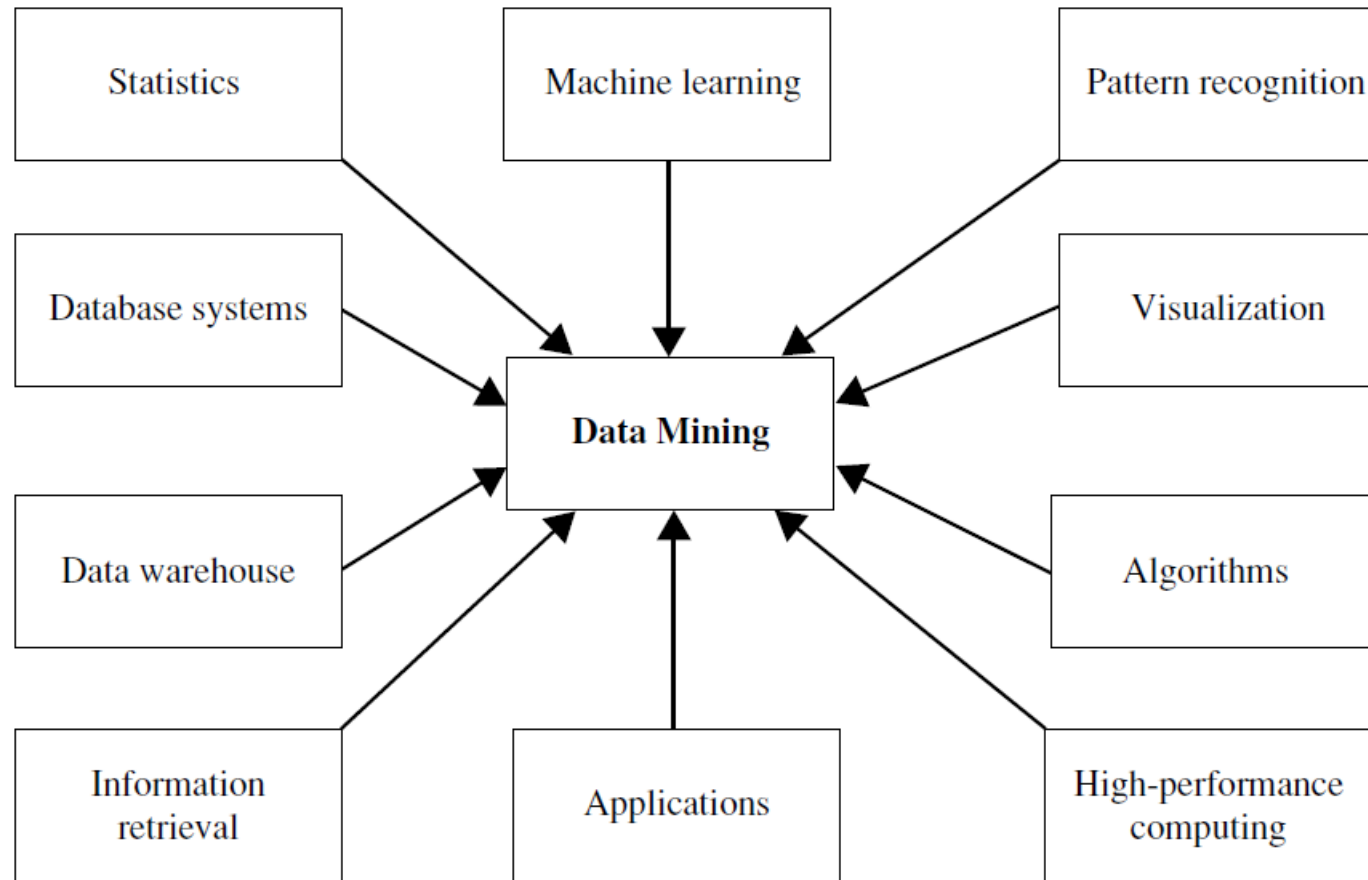
- Several **objective measures of pattern interestingness** exist.
 - E.g, An objective measure for **association** rules of the form $X \Rightarrow Y$ is rule **support**, representing the percentage of transactions from a transaction database that the given rule satisfies.
 - Another objective measure for **association** rules is **confidence**, which assesses the degree of certainty of the detected association.
- Although objective measures help identify interesting patterns, they are often insufficient unless combined with subjective measures that reflect a particular user's needs and interests.
- **Subjective interestingness measures** are based on user beliefs in the data.

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- *Can a data mining system generate all of the interesting patterns?*—refers to the **completeness** of a data mining algorithm.
- *Can a data mining system generate only interesting patterns?*— is an **optimization** problem in data mining.

Which Technologies are used?



Data mining adopts techniques from many domains

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- A **statistical model** is a set of mathematical functions that describe the behavior of the objects in a target class in terms of random variables and their associated probability distributions.
- **Machine learning** investigates how computers can learn (or improve their performance) based on data. A main research area is for computer programs to **automatically** learn to recognize complex patterns and make intelligent decisions based on data.
 - **Supervised learning**- The supervision in the learning comes from the labeled examples in the training data set. E.g., Classification tasks
 - **Unsupervised learning**- The learning process is unsupervised since the input examples are not class labeled. Typically, clustering is used to discover classes within the data.
 - **Semi-supervised learning** is a class of machine learning techniques that make use of both labeled and unlabeled examples when learning a model.
 - **Active learning** is a machine learning approach that lets users play an active role in the learning process.
 - **Transfer learning** is a technique in machine learning where a model trained on one task is reused or adapted to a new, but related task. It aims to extract knowledge from one or more source tasks and apply the knowledge to a target task. It leverages knowledge gained from solving one problem and applies it to another, which is particularly useful when the new task has limited data available for training.

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- Data warehouse systems provide multidimensional data analysis capabilities, collectively referred to as **online analytical processing**.
- **Information retrieval (IR)** is the science of searching for documents or information in documents.
 - The typical approaches in information retrieval adopt probabilistic models.
 - The document's **language model** is the probability density function that generates the bag of words in the document.

Major Issues in Data Mining



- Mining methodology,
- User interaction,
- Efficiency and scalability,
- Diversity of data types
- Data mining and society



End of Lecture!

hsumyatmo@ucsy.edu.mm